

The Body Rescue Mission

Bonus · Human Physiology

Starring The Micro-Medics 👩‍⚕️

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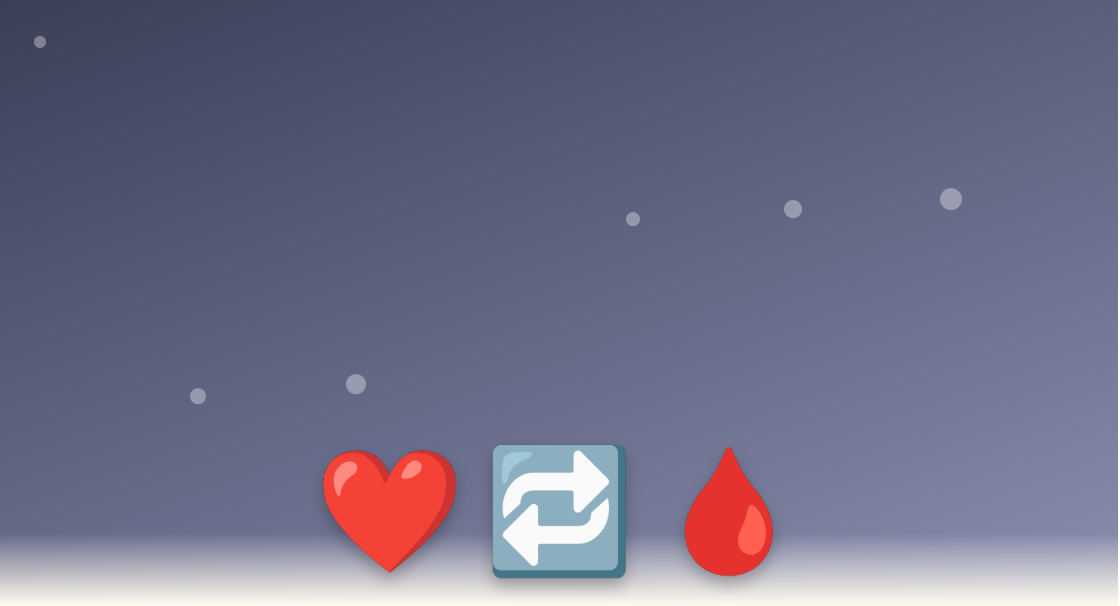


Alarm in the Body

An alarm blared inside a human body: "Energy levels low. Oxygen delivery weak!" A team of tiny medical explorers, the Micro-Medics, shrank into a capsule and set off. "Our mission," said their leader, Dr. Asha, "is to check five **organ systems**: circulatory, respiratory, digestive, nervous and excretory. Every one must work together."

💡 **REMEMBER**

Organs that work together for a common function form an organ system.



Stop 1: The Heart

The capsule rode the bloodstream into the **heart**, with four chambers: two **atria** and two **ventricles**. The **SA node**, the **pacemaker**, fired about 72 times a minute. "Blood passes through the heart twice in one round — **double circulation**," said Dr. Asha. Each beat pushed about 70 mL — the **stroke volume**.

💡 **REMEMBER**

Cardiac output = stroke volume × heart rate ≈ 5 litres per minute. One cardiac cycle lasts about 0.8 s.



Stop 2: The Lungs

Next were the **lungs**, branching into millions of tiny air sacs, the **alveoli** — the site of **gas exchange**. Oxygen diffused into the blood and joined **haemoglobin** in red blood cells as **oxyhaemoglobin**. CO_2 moved out, mostly carried as **bicarbonate**. Breathing rhythm is set by the **respiratory rhythm centre** in the **medulla**.

💡 **REMEMBER**

About 97 percent of O_2 is carried by RBCs; about 70 percent of CO_2 is carried as bicarbonate.



Stop 3: The Stomach

In the **stomach**, **gastric glands** poured out **hydrochloric acid** and **pepsinogen**. The acid converted pepsinogen into **pepsin**, which began digesting proteins. Earlier, in the mouth, **salivary amylase** had started breaking starch. "Food here becomes an acidic paste called **chyme**," Dr. Asha explained.

💡 **REMEMBER**

Mucus and bicarbonates protect the stomach lining from HCl.



Stop 4: The Intestine

In the **small intestine**, **bile** from the liver **emulsified** fats into tiny droplets, and **pancreatic juice** brought trypsin, amylase and lipase. Finger-like **villi** covered the walls, giving a huge surface for **absorption**. Simple sugars, amino acids and fatty acids passed into the blood and lymph. Nutrients were on their way to every cell!

💡 **REMEMBER**

Most digestion and absorption happen in the small intestine; water is mainly absorbed in the large intestine.



Stop 5: The Brain

The capsule zipped up to the **brain**, the control centre. Billions of **neurons** sent electrical **nerve impulses**, passing messages across gaps called **synapses** using **neurotransmitters**. The **cerebrum** handled thinking, the **cerebellum** kept balance, and the **medulla** controlled breathing and heartbeat.

💡 **REMEMBER**

The resting membrane of a neuron is negative inside; an impulse is a wave of depolarisation.



Stop 6: The Kidneys

Finally they reached the **kidneys**, each with about a million **nephrons**. In the **glomerulus**, blood was filtered — **glomerular filtration** — about 125 mL per minute, or 180 litres a day! Nearly 99 percent was taken back by **reabsorption**. Waste **urea** left as urine. **ADH** helped save water when the body was thirsty.

💡 **REMEMBER**

Kidneys are the main organs of excretion; urea is formed in the liver.



All Systems Go!

With every system checked, oxygen reached the cells, nutrients were absorbed, messages flowed and waste was removed. The alarm turned green: "Body at full health." Dr. Asha smiled. "No system works alone. The heart needs the lungs, the cells need the gut, and the kidneys keep the blood clean. Mission complete!"

💡 **REMEMBER**

Coordination between organ systems keeps the internal environment stable — homeostasis.



KEY TERMS

Words you met in this story

Tap a word to jump to its page. Can you explain each one?

organ systems p.1 · heart p.2 · atria p.2 · ventricles p.2 · SA node p.2 · pacemaker p.2 · double circulation p.2 · stroke volume p.2 · lungs p.3 · alveoli p.3 · gas exchange p.3 · haemoglobin p.3 · oxyhaemoglobin p.3 · bicarbonate p.3 · respiratory rhythm centre p.3 · medulla p.3 · stomach p.4 · gastric glands p.4 · hydrochloric acid p.4 · pepsinogen p.4 · pepsin p.4 · salivary amylase p.4 · chyme p.4 · small intestine p.5 · bile p.5 · emulsified p.5 · pancreatic juice p.5 · villi p.5 · absorption p.5 · brain p.6 · neurons p.6 · nerve impulses p.6 · synapses p.6 · neurotransmitters p.6 · cerebrum p.6 · cerebellum p.6 · kidneys p.7 · nephrons p.7 · glomerulus p.7 · glomerular filtration p.7 · reabsorption p.7 · urea p.7 · ADH p.7



QUICK QUIZ

Do you remember?

1. Which system transports oxygen and nutrients?

- Digestive
- Circulatory
- Excretory
- Nervous

2. Where does gas exchange occur in the lungs?

- Trachea
- Bronchi
- Alveoli
- Diaphragm

3. Which organ filters waste from the blood?

- Liver
- Kidney
- Stomach
- Heart

4. The pacemaker of the heart is the:

- AV node
- SA node
- Bundle of His
- Medulla

5. Fats are emulsified by:

- Pepsin
- Bile
- Saliva
- HCl

Answers: 1-Circulatory · 2-Alveoli · 3-Kidney · 4-SA node · 5-Bile